FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Advanced)-2021

PAPER – 2

TEST DATE: 20-12-2020

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total **54** questions.
- Each subject (PCM) has **18** questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section – B & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks **: –2** In all other cases

Section-B (07 – 12, 25 – 30, 43– 48) *contains* ***18 Numerical*** *based questions with* ***Single digit integer*** *as answer, ranging from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-C (13 – 18, 31 – 36, 49– 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

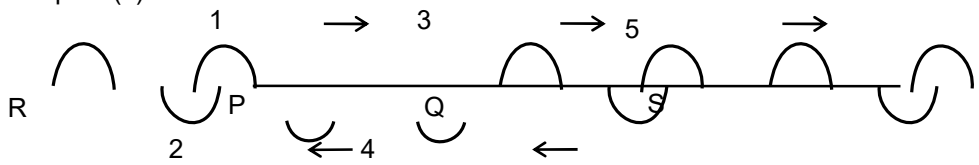
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **06** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. There are three strings RP, PQ, QS as shown in the figure. Their mass and lengths are RP = (0.1kg, 2m), PQ = (0.2kg, 3m), QS = (0.15kg, 4m) respectively. All the strings are under same tension. Wave-1 is incident at P. It is partly reflected (Wave-2) and partly transmitted (wave-3). Now wave-3 is incident at Q. It is again partly transmitted (wave-5) and partly reflected (wave-4). Choose the correct option(s).



(A) Phase difference between the wave-1 and the wave-2 is π.

(B) Phase difference between the wave-3 and the wave-4 is zero .

(C) $\frac{A_3}{A_1} = \frac{2\sqrt{3}}{2+\sqrt{3}}$

(D) $\frac{A_4}{A_3} = \frac{1}{7}$

2. A string of mass m is fixed at both of its ends. The fundamental mode of string is excited and it has an angular frequency ω and maximum displacement amplitude A. Then :

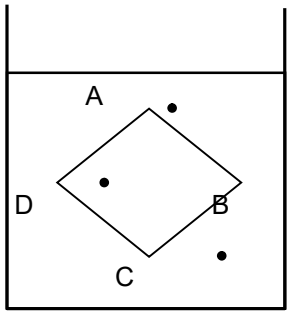
(A) The maximum kinetic energy of the string is $E_k = \frac{1}{4}mA^2\omega^2$

(B) The maximum kinetic energy of the string is $E_k = \frac{1}{2}mA^2\omega^2$

(C) The mean kinetic energy of the string averaged over tone periodic time is $< E_k >= \frac{1}{4}mA^2\omega^2$

(D) The mean kinetic energy of the string averaged over tone periodic time is $< E_k >= \frac{1}{8}mA^2\omega^2$

3. The figure shows a container filled with a liquid of density ρ. Four points A, B, C and D lie on the vertices of a vertical square. Points A and C lie on a vertical line and points B and D lie on a horizontal line. Choose the correct statement (s) about the pressure at the four points



(A) $P_D = P_B$

(B) $P_A < P_B = P_D < P_C$

(C) $P_D = P_B = \frac{P_C - P_A}{2}$

(D) $P_D = P_B = \frac{P_C + P_A}{2}$

4. A body suspended from a spring balance is kept in a satellite. The reading of the balance is W_1 when the satellite goes in an orbit of radius R and is W_2 when it goes in an orbit of radius 2R. Mark the incorrect options:
   (A) $W_1=W_2$
   (B) $W_1<W_2$
   (C) $W_1>W_2$
   (D) $W_1\neq W_2$

5. Consider (i) a vernier callipers in which each 1 cm on the main scale is divided into 8 equal divisions and (ii) a screw gauge with 100 divisions on its circular scale. In the vernier callipers 5 divisions of the vernier scale coincides with 4 divisions on the main scale and in the screw gauge, one complete rotation of the circular scale moves it by two divisions on the linear scale. Now, which of the following statements is/are correct?
   (A) If the pitch of the screw gauge is twice the least count of the vernier callipers, the least count of the screw gauge is 0.01 mm.
   (B) If the pitch of the screw gauge is twice the least count of the vernier callipers, the least count of the screw gauge is 0.005 mm.
   (C) If the least count of the linear scale of the screw gauge is twice the least count of the vernier callipers, the least count of the screw gauge is 0.01 mm.
   (D) If the least count of the linear scale of the screw gauge is twice the least count of the vernier callipers, the least count of the screw gauge is 0.005 mm.

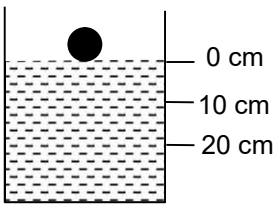
6. Two spherical bodies, each of mass M, are kept with their centres at a distance 2L apart. A particle of mass m is projected from the mid-point of the line joining their centres and perpendicular to it. The gravitational constant is G. Which of the following statements is/are true?
   (A) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $4\sqrt{\frac{GM}{L}}$
   (B) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $2\sqrt{\frac{GM}{L}}$
   (C) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $\sqrt{\frac{2GM}{L}}$
   (D) The total mechanical energy of the mass m remains constant.

SECTION – B
(Single digit integer type)

This section contains **06** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

7. A cylindrical vessel open at the top is 20 cm high and 10 cm in diameter. A circular hole whose cross-sectional area 1 cm^2 is cut at the centre of the bottom of the vessel. Water flows from a tube above it into the vessel at the rate 100 cm^3s^{-1}. Find the height (in cm) of water in the vessel under steady state. (Take g = 10 m/s^2)

8. A spherical ball of density ρ and radius 0.003m is dropped into a tube containing a viscous fluid filled up to the 0 cm mark as shown in the figure. Viscosity of the fluid = 1.260 N-s.m^{-2} and its density $\rho_L = \rho/2$ = 1260 kg.m^{-3}. Assume the ball attains the terminal speed by the 10 cm mark. Find the time taken (in sec) by the ball to traverse the distance between the 10 cm and 20 cm mark. (Take g = 10 ms^{-2})



9. A fish is near the centre of a spherical bowl filled with water (μ = 4/3). A child stands in air at a distance 2R (R is the radius of curvature of the sphere) from the centre of the bowl. At what distance (in m) from the centre would the child nose appear to the fish situated at the centre. (Given R = $\frac{7}{3}$m)

10. An object is placed at a distance of 15 cm from a convex lens of focal length 10 cm. On the other side of the lens, a convex mirror is placed at its focus such that the image formed by the combination coincides with the object itself. If the focal length of the convex mirror is 5n cm. Find the value of n.

11. Two particles of a medium disturbed by the wave propagation are located at $x_1 = 0$ and $x_2 = 1$cm. The respective displacements (in cm) of the particles can be given by the equations
$y_1 = 2 \sin 3\pi t$
$y_2 = 2\sin\left(3\pi t - \frac{\pi}{8}\right)$
The wave velocity is 3k cm/sec. Find the value of k.

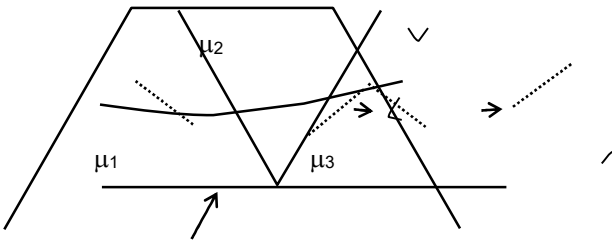
12. An ionization counter is used to investigate the disintegration rate of a certain radioactive sample. At the start of the experiment , the counter gives 141 pulses in 20 seconds . After 3 days it gives 100 pulses in 20 seconds . Find the half life of the sample in days (Take $\sqrt{2}$ = 1.41).

SECTION – C
(Numerical Answer Type)

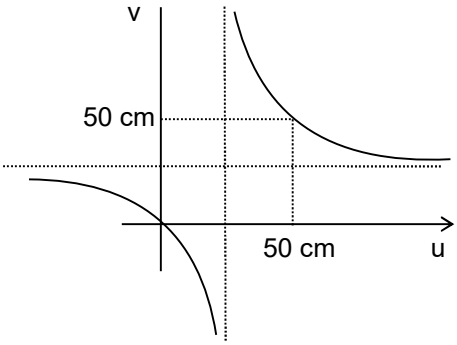
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. In a certain hypothetical radioactive decay process, species A decays into species B and species B decays into species C according to the given reactions.
A $\rightarrow$ 2B + particles + energy
B $\rightarrow$ 3C + particles + energy
The decay constant for species A is $\lambda_1 = 1$ sec^{-1} and that for species B is $\lambda_2 = 100$ sec^{-1}. Initially 10^4 moles of species of A were present, while there were none of B and C. It was found that species B reaches its maximum number at a time $t_0 = 3\ell n(5)$ sec. Calculate the value of maximum number of moles of B.

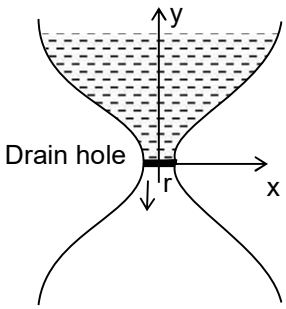
14. A ray of light enters at grazing angle of incidence into an assembly of three isosceles right-angled prisms having refractive indices $\mu_1 = \sqrt{2}$, $\mu_2 = \sqrt{x}$ and $\mu_3 = \sqrt{3}$. If finally emergent light ray also emerges at grazing angle then calculate x :



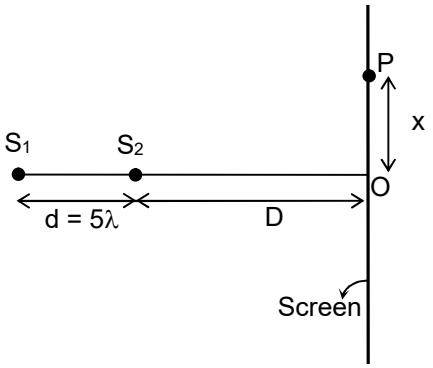
15. **u-v** graph for a convex mirror is shown in the figure. (here |u| denotes the object distance and |v| is the image distance). Find the lateral magnification of a real object placed at a distance 25 cm from the pole of mirror.



16. The shape of an ancient water clock jar is such that the water level descends at a constant rate at all times. The water level falls by λ m/s and the shape of the jar is given by $y \alpha x^n$. Find the value of $\left(\frac{3n}{5}\right)$, if the radius r of the drain hole can be assumed to be very small.



17. Two coherent point sources S_1 and S_2 are placed on a line perpendicular to the screen as shown in the figure. The wavelength of the light emitted by the sources is 'λ'. The distance between the two sources is $d = 5\lambda$. The distance of S_2 from the screen is D (>>λ). Find the minimum (non zero) distance x (in meter) of a point P on the screen from the central point O at which maxima is obtained. (Given D = 1.6 m)



18. The energy of a system as a function of time t is given as $E(t) = A^2 \exp(-\alpha t)$, where $\alpha = 0.2\ s^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%, find the maximum percentage error in the value of E(t) at t = 4 s.

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **06** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which of the following is/are square planar in shape?
(A) $[PdCl_4]^{2-}$
(B) $[Ni(CN)_4]^{2-}$
(C) $[Pd(CN)_4]^{2-}$
(D) $[NiCl_4]^{2-}$

20. The ratio of average kinetic energy of triatomic gas and monoatomic gas at same temperature is/are? [Neglect vibrational degree of freedom]
(A) 2
(B) $\frac{3}{2}$
(C) $\frac{5}{6}$
(D) $\frac{5}{3}$

21. The mean free path of the molecule of a certain gas at 300 K is 3.5×10^{-5} m. The collision diameter of the molecule is 0.25 nm which of the following is/are correct?
(A) Pressure $= 4.2\times10^{-3}$ atm
(B) Number of molecules per unit volume of gas is 1.03×10^{23} m^{-3}
(C) Pressure $= 4.26\times10^{2}$ Pa
(D) A and B only

22. Which of the following statement(s) for crystal having Frankel defect is/are correct?
(A) Frenkel defects are observed where the difference in sizes of cations and anions is large.
(B) Pure/Ideal alkali metal halides do not have Frenkel defect
(C) The density of crystals having Frenkel defect is lesser than that of a pure/ideal crystal.
(D) An ionic crystal may have Frenkel defect along with Schottky defect.

23. Which of the following statement(s) is/are correct?
(A) The co-ordination number of each type of ion in CsCl crystal is 8.
(B) Rhombohedral crystals have only one Bravais lattice.
(C) In cubic close packed structures effective number of octahedral voids and effective number of spheres per unit cell is same.
(D) The hcp and ccp structure for a given element have same packing fraction.

24. Which of the following are correct?
(A) Adsorption is accompanied by decrease in enthalpy.
(B) Fog is aerosol
(C) The critical micellization concentration (CMC) is the concentration at which micellization begins
(D) Purple of cassius is colloidal solution of gold

SECTION – B
(Single digit integer type)

This section contains **06** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

25. 1.1 gm $CoCl_3.6NH_3$ was dissolved in 100 gm of water. The freezing point of solution was – 0.383°C. How many moles of solute particles exist in solution for each mole of solute introduced? [Given K_f of $H_2O = 1.86$ K mol^{-1} kg , molar mass of Co = 58.9, Cl = 35.5, NH_3 = 17] (Assume complete ionization of solute and assume the solute to be non-volatile).

26. If the number of moles of electrons lost by one mole of ferrous thiosulphate in its reaction with chlorine water are x, then the value of 'x' is……..

27. $P_4 + OH^- \longrightarrow x + y$

If sum of the oxidation number of phosphorus in x and y is equal to $(-z)$ then z is

28. Which of the following halides are not colourless (white)
AgI, PbI_2, Hg_2I_2, HgI_2 , AgCl, $PbCl_2$, Hg_2Cl_2, AgBr

29. The unit cell of aluminium (Molar mass of Al = 27.0 gm/mol) is a cube with edge length 404 pm. The density of aluminium is 2.70 gm/cm^3. What are the effective number of atoms (or Rank of unit cell) per unit cell? $\left(N_{av} = 6.022 \times 10^{23}\right)$

30. A compound xy has a rock salt type structure with x : y = 1 : 1 the formula weight of xy is $6.023\,A$ amu and the closest x – y distance is $A^{1/3}$ nm. Find the density of lattice (in kg/m^3) $\left(N_{av} = 6.023 \times 10^{23}\right)$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. Dry air was passed successively through a solution of 6 gm of a solute in 185 gm of water and then through pure water. The loss in weight of solution was 2.6 gm and that of pure solvent was 0.4 gm. What is the molecular mass of the solute (g/mole)? (Assume solute to be non-volatile and non-electrolyte)

32. A non-volatile and non-electrolyte solute is dissolved in 55.4 cc of benzene (density 0.889 gm cm^{-3}) at room temperature, vapour pressure of this solution is 95.85 mm Hg while that of pure benzene is 100 mm Hg. If the freezing temperature of this solution is 0.60 degree lower than that of benzene, what is the value of molal freezing point depression constant of benzene. (in K kg mol^{-1})

33. For a saturated solution of AgCl at 25°C, specific conductance is 3.32×10^{-6} ohm^{-1} cm^{-1} and that of water used for preparing the solution was 1.5×10^{-6} ohm^{-1} cm^{-1}. If the solubility product of AgCl is $x \times 10^{-10}$, then the value of x is…..
[Given $\lambda^{\infty}_{eqv}(AgCl) = 138.3$ ohm^{-1} cm^2 eqv^{-1}]

34. EMF of the following cell is 0.58 V at 298 K

$Pt\,|\,H_2\,(1\,atm)\,|\,H^+\,(PH = x)\,||\,KCl(1\,N)\,|\,Hg_2Cl_2\,(s)\,|\,Hg$

Calculate pH of the anode compartment.

Given $E^\circ_{Cl^-|Hg_2Cl_2|Hg} = 0.28\ V$

35. Calculate the free energy change at 298 K for the reaction

$Br_2\,(\ell) + Cl_2\,(g) \longrightarrow 2BrCl(g)$

For the reaction $\Delta H^\circ = 25.7\,kJ$ and the standard entropies of $Br_2\,(\ell)$, $Cl_2\,(g)$ and $BrCl(g)$ at the 298 K are 151.3, 221.0, 238.9 J mol^{-1} K^{-1} respectively. If ΔG° of the reaction is – x Joule then $\frac{x}{100}$ is.....

36. A compound AB crystallizes as CsCl lattice with unit cell edge length of 485 pm. If the radius of B^- is 235 Pm, then what will be the radius of A^+ (in pm)

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **06** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If the value of $^{2000}C_2 + {}^{2000}C_5 + {}^{2000}C_8 + {}^{2000}C_{11} ..... {}^{2000}C_{2000}$ is $\frac{2^{k_1}-1}{k_2}$, then

(A) $k_1 - 666k_2 = 1$
(B) $k_1 - 666k_2 = 2$
(C) $k_1 - 13^{k_2} = 197$
(D) $k_1 - 13^{k_2} = -197$

38. The polynomial $1 - x + x^2 - x^3 + ... - x^{17}$ may be written in the form $a_0 + a_1y + a_2y^2 + ... + a_{16}y^{16} + a_{17}y^{17}$ where $y = x + 1$ and all a_i's are constant, then

(A) $a_2 = -{}^{18}C_3$
(B) $a_1 = {}^{18}C_2$
(C) $a_2 = {}^{18}C_3$
(D) $a_1 = -{}^{18}C_2$

39. Let S be the set of all non zero real number 'a' such that the quadratic equation $ax^2 + (a - 1)x + (a - 2) = 0$ has two distinct real roots α and β satisfying the inequality $|\alpha - \beta| < 2$. Then which of the following interval(s) is/are a subset of S?

(A) $\left(1+\frac{2}{\sqrt{3}}, \infty\right)$
(B) $\left(1-\frac{2}{\sqrt{3}}, 1\right)$
(C) $\left(1, 1+\frac{2}{\sqrt{3}}\right)$
(D) $\left(1-\frac{2}{\sqrt{3}}, -\frac{1}{7}\right)$

40. Given an alphabet with three letter a, b, c. Let the number of words of n letters which contain an even number of a's are denoted by f(n), then

(A) f(3) = 14
(B) f(4) = 41
(C) f(3) = 13
(D) f(4) = 40

41. Let a line L passing through a point (–1, –2, 3) is perpendicular to two lines.
$\frac{x-1}{2}=\frac{y-1}{1}=\frac{z-1}{4}$ and $\frac{x-0}{1}=\frac{y-2}{3}=\frac{z-1}{2}$.
If perpendicular distance of a point on line L from the plane x + y + z – 3 = 0 is $2\sqrt{3}$, then the point(s) is/are
(A) (–3, –2, 4)
(B) (–5, –2, 5)
(C) (17, –2, –6)
(D) (–7, –2, 6)

42. There are 4 randomly chosen non-negative integer x, y, z, ω are found to satisfy the equation x + y + z + ω = 20, then events are defined
E_1 = all four numbers x, y, z, ω are odd number
E_2 = x is a prime number greater than 10
(A) $P(E_1)=\frac{15}{161}$
(B) $P(E_1)=\frac{20}{161}$
(C) $P(E_2)=\frac{208}{1771}$
(D) $P(E_2)=\frac{104}{1771}$

SECTION – B
(Single digit integer type)

This section contains **06** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

43. The number of real values of x for which $\frac{8^x+27^x}{12^x+18^x}=\frac{7}{6}$

44. Let z_1, z_2 be complex numbers with $|z_1| = |z_2| = 1$, then min $(|z_1 + 1| + |z_2 + 1| + |z_1z_2 + 1|)$ is equal to

45. Let z be a complex number such that $z \in C / R$ (not a real number) and $\frac{1+z+z^2}{1-z+z^2}\in R$, then |z| is equal to

46. In a tetrahedron ABCD, E and F are the centroid of face BCD, ACB. Then the ratio of $\frac{\text{Volume of tetrahedron ABCD}}{\text{Volume of tetrahedron BCEF}}$ is

47. Two lines are formed by intersection of plane 2x + 3y + 4z – 1 = 0 with the planes x + y + z – 3 = 0 and x + y + z + 3 = 0, then if the shortest distance between both the lines is $\sqrt{(29)(6^k)}$, then k is equal to

48. If $p(x)=\begin{vmatrix}1 & 1 & 1\\ (x+2) & (x+3) & (x+4)\\ (x+2)^2 & (x+3)^2 & (x+4)^2\end{vmatrix}$, then value of $\frac{p(3)-p(4)}{p(5)}$ is equal to

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Letters of the word "MOTIVATION" are arranged randomly. If the probability of these such words contains the word "IIT" always is 'p', then $\frac{1}{p}$ is equal to

50. For any sequence of real numbers A = {a_1 a_2, a_3, a_4 .....}, define ΔA to be the sequence {$a_2 - a_1$, $a_3 - a_2$, $a_4 - a_3$ .....}. Suppose that all the term of sequence $\Delta(\Delta A)$ are 1, and $a_{51} = a_{41} = 0$, then the value of a_{11} is equal to

51. The value of expression $\frac{3}{1!+2!+3!}+\frac{4}{2!+3!+4!}.....\frac{2021}{2019!+2020!+2021!} = \frac{1}{k_1}-\frac{1}{k_2!}$, then $k_2 - k_1$ is

52. Consider all 5 digit numbers; from these numbers 'a' numbers have the property that product of their digits is 25 and 'b' numbers have the property that product of their digits is 15, then $\frac{a}{b}$ is

53. In the last over of a cricket match, Virat is on strike end and 12 runs are required to win the match. Assume that there isn't any strike change and no wide or no ball is thrown in over, then the number of ways in which Virat will score 12 runs (all the 6 balls were played by Virat and exactly 12 runs were scored, overthrow was not allowed)

54. Let consider matrix A of 3×3 order where A is $\begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ and I is a 3×3 scalar matrix, then if $A^{-2} = 10A^2 + aA + bI$, then 'a + b' is equal to